INCH-POUND

MIL-PRF-38230C(USAF)
1 October 1997
SUPERSEDING
MIL-T-38230B(USAF)
15 April 1988

PERFORMANCE SPECIFICATION

TRANSMITTER, PRESSURE, OIL

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers oil pressure transmitters.

1.2 Classification. Transmitters will be of the following type, as specified (see 6.2):

TRU-66/A 100 percent (200 psi) overpressure

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. These lists do not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet the requirements specified in the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of these documents are those listed in the specific issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, 3001 Staff Drive, Suite 1AE1-101A, Tinker AFB, OK 73145-3036, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620
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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-25435	Transmitters, Pressure, Aircraft, General Specification for
MIL-I-25437	Indicators, Aircraft, Pressure, Voltage Ratio Type, General Specification for
MIL-I-25438	Indicator, Pressure Type MO-2
MIL-I-27189	Indicator, Pressure, Oil
MIL-I-38133	Indicator, Pressure, Oil

(Unless otherwise indicated, copies of the above specifications are available from the Defense Automated Printing Service, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of the documents which have been adopted by the DoD are those listed in the specific issue of the DoDISS cited in the solicitation. Unless otherwise specified, the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE/AMERICAN SOCIETY FOR QUALITY (ANSI/ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD-adopted)

(Application for copies should be addressed to American Society for Quality, P.O. Box 3066, Milwaukee, WI 53201-3066, or to the American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The transmitters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 General requirements. The requirements of MIL-PRF-25435 apply as requirements of this specification with the exceptions and additions specified herein.

3.3 Workmanship. The transmitter, including all parts, shall be constructed in accordance with commonly accepted industrial workmanship standards.

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3.4 Interface.

3.4.1 Pressure indicators. The transmitter shall sense the differential of two pressures, convert the differential pressure to an electrical signal, and provide the unamplified signal to an indicator conforming to MIL-I-25437, MIL-I-25438, MIL-I-27189, or MIL-I-38133. External wiring between the transmitter and indicator shall be as shown in figure 1.

3.5 Performance.

3.5.1 Transmitter error tolerances. The transmitter shall meet error tolerances when performing the scale error, friction error, position error and vibration tests in MIL-PRF-25435. The test points and error tolerances are specified in tables I and II. The transmitter shall operate within these tolerances when connected to a standard indicating unit which has been calibrated to a maximum readout error of ±0.25 percent. The transmitter shall be connected to the standard indicating unit using the standard ratio transformer circuit in figure 2.

Table I. Scale error

Test points (psi)	Scale error test tolerances (±psi)		
	Note 1	Note 2	Note 3
0	1.0	1.75	2.0
10	1.0	1.75	2.0
20	1.0	1.75	2.75
30	1.75	2.25	2.75
40	1.75	2.25	2.75
50	1.75	2.25	2.75
60	1.75	2.25	2.75
70	1.75	2.25	2.75
80	1.75	2.25	2.75
90	2.0	2.5	3.0
100	2.0	2.5	3.0

Note 1: Tolerance after the scale and friction error test at ambient temperature in MIL-PRF-25435.
Note 2: Tolerance after endurance (10,000 cycles), humidity, power, and overpressure tests.
Note 3: Tolerance after vibration, low temperature, high temperature, endurance (60,000 cycles), temperature-altitude, and low temperature exposure tests. Also, the tolerance after the transmitter has been subjected to the 80-hour high temperature exposure test and is returned to ambient temperature for minimum of 4 hours.

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Table II. Friction, position, and vibration error

Test	Test tolerance (± psi)
Friction error (Note 1)	
Ambient temperature	0.5
Low temperature	1.5
High temperature	0.5
Endurance (10,000 cycles)	0.5
Endurance (60,000 cycles)	0.5
Temperature-altitude	1.5
Low temperature exposure	1.5
High temperature exposure	0.5
Humidity	0.5
Power	0.5
Overpressure	0.5
Position error at 50 psi test point only	0.5
Vibration error (while vibrating)	
Pointer oscillation	0.5
Pointer variation	0.5
Note 1: Friction error will be tested at test points in table I.	

3.5.2 Overpressure. The TRU-66/A transmitter shall withstand a 100% (200 psi) overpressure for a maximum of 10 minutes without being damaged.

3.5.3 Wire length. The transmitter shall operate as specified with a No. 20 AWG stranded copper wire between the transmitter and indicator up to 300 feet long.

3.5.4 Weight. The weight of the transmitter shall be 1.0 pound or less.

3.5.5 Reliability. The transmitter shall have a minimum mean time between failures (MTBF) of 2,000 hours.

3.5.6 Operating life span. The transmitter shall have a minimum operating life span of 3,000 hours.

3.6 Item identification. The transmitter shall be permanently and legibly marked with the following information (see 6.2).

a. Manufacturer’s name

b. Manufacturer’s CAGE code

c. Manufacturer’s part number

d. Manufacturer’s lot number

e. National stock number

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f. Transmitter type

g. Date of manufacture

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification (see 4.2)

b. Conformance (see 4.3)

4.2 Qualification. Qualification shall be performed on five transmitters of the type procured. Four of the transmitters shall be subjected to all tests in 4.5 of this document and all tests in 4.6 of MIL-PRF-25435 except the fungus test. The remaining transmitter shall be subjected to the fungus test in MIL-PRF-25435.

4.3 Conformance. A sample of transmitters shall be selected in accordance with ANSI/ASQ Z1.4 and shall be subjected to the dielectric strength, scale and friction error, position error, leakage, vibration, and low temperature tests specified in MIL-PRF-25435. The sample of transmitters shall also be subjected to the examinations in MIL-PRF-25435 and 4.5.1 of this document.

4.4 Requirements cross-reference matrix. Table III provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

Table III. Requirements cross-reference matrix

Requirement	Verification
3.1	4.2
3.2	4.2
3.3	4.5.1
3.4.1	4.5.2, 4.5.6
3.5.1	4.5.2
3.5.2	4.5.3
3.5.3	4.5.2
3.5.4	4.5.1
3.5.5	4.5.4
3.5.6	4.5.5
3.6	4.5.1

4.5 Tests.

4.5.1 Examination. The transmitter shall be examined for compliance with requirements for workmanship, weight, and markings (item identification).

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4.5.2 Transmitter error. The transmitter shall be subjected to the tests specified in the notes in table I, and then shall meet the tolerances specified for scale, friction, position, and vibration errors in tables I and II. The pointer oscillation and variation tolerances for the vibration error in table II shall be verified while conducting the vibration test. The transmitter shall meet these tolerances when connected to the standard indicating unit through 300 feet of No. 20 AWG stranded copper wire.

4.5.2.1 Example. To verify that the transmitter meets the scale error tolerance after performing the overpressure test, refer to the Note 2 column and select a test point. If 20 psi is selected, the tolerance is ±1.75 psi. The standard indicating unit shall read 18.25 to 21.75 psi when performing the scale error test. The friction error allowed for this test is ±0.5 psi as shown in table II. Friction error is the difference in readings observed before and after tapping the transmitter while applying the test pressure of 20 psi.

4.5.3 Overpressure. The TRU-66/A transmitter shall be subjected to the overpressure test in MIL-PRF-25435 and then shall meet the scale error tolerances in table I of this document.

4.5.4 Reliability. If specified in the acquisition document (see 6.2), the reliability test shall be performed.

4.5.4.1 Preparation. The transmitter shall be mounted directly to a vibration exciter without vibration isolators and shall be placed in a chamber for temperature cycling. The transmitter shall be connected to the standard indicating unit and pressure and electrical power sources. Pressure shall be applied to the transmitter at a rate of 3 cycles per minute to obtain a reading of 0 to 85% of full scale and return to 0 on the standard indicating unit during the reliability test.

4.5.4.2 Test. Rated power (see figure 1) and pressure shall be applied to the transmitter for 4 hours. The transmitter shall be subjected to random vibration for a minimum of 10 minutes during each hour when power is applied. The level of the acceleration power spectral density (APSD) shall be: 0.013 g^2/Hz at 15 Hz; increase at a rate of 4dB per octave to 0.7 g^2/Hz at 300 Hz; maintain 0.7 g^2/Hz from 300 to 1,000 Hz; and decrease at a rate of 6dB per octave to 0.176 g^2/Hz at 2,000 Hz. Temperature cycling shall be conducted as follows: the transmitter shall be brought to -65°F and maintained at that temperature for 4 hours; increase the transmitter temperature to 330°F, at a minimum rate of 9°F per minute, and maintain at that temperature for 4 hours; decrease the transmitter temperature to -65°F , at a minimum rate of 9°F per minute, and maintain at that temperature for 4 hours. Continue this sequence during the reliability test. The electrical power shall be turned off for 10 minutes after each 4 hours with power applied to the transmitter. The transmitter shall be subjected to the scale error test in MIL-PRF-25435 after each 250 hours of testing and shall not exceed 150% of the ambient temperature tolerance in table I. Transmitter malfunctions shall be recorded and repairs made. The transmitters shall not be recalibrated during the reliability test. The reliability test shall demonstrate a transmitter MTBF of 2,000 hours with a confidence level of 90% and a discrimination ratio of 1.5.

4.5.5 Operating life span. After satisfactory completion of the reliability test, and if specified in the acquisition document (see 6.2) an operating life span of 3,000 hours shall be demonstrated by continuing the reliability test, and repairing the transmitter when failures occur, until the transmitter has operated 3,000 hours.

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4.5.6 Compatibility. The transmitter shall be connected to and tested with indicators specified in MIL-I-25437, MIL-I-25438, MIL-I-27189, and MIL-I-38133. The transmitter and indicators shall be connected as shown in figure 1. The scale error test in MIL-PRF-25435 shall be performed at ambient temperature. The tolerance at each test point in table I shall be the sum of the tolerances in table I of this document and table I of MIL-I-25438 at ambient temperature.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The transmitters covered by this specification are intended for use with variable-reluctance-type indicators conforming to MIL-I-25437, MIL-I-25438, MIL-I-27189, and MIL-I-38133 in oil pressure indicating systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. Item identification (see 3.6).

d. Type of transmitter required (see 1.2).

e. Data to be submitted with the qualification samples.

f. If a reliability test is required (see 4.5.4).

g. If an operating life span test is required (see 4.5.5).

h. Packaging requirements (see 5.1).

6.3 Qualification. The attention of the contractors is called to the requirements with respect to products requiring qualification. Awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the Qualified Products List (QPL No. 38230)

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whether or not such products have actually been listed by that date. In order that the manufacturers may be eligible to be awarded contracts or purchase orders for the products covered by this specification, they are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification. Information pertaining to qualification of products may be obtained from Oklahoma City Air Logistics Center/TICLA, 3001 Staff Drive, Suite 1AE1-101, Tinker AFB, OK 73145-3036.

6.4 Subject term (key word) listing.

TRU-66/A
Variable reluctance
0-100 PSI

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 71
(Project No. 6620-F651)

Agent
Air Force - 99

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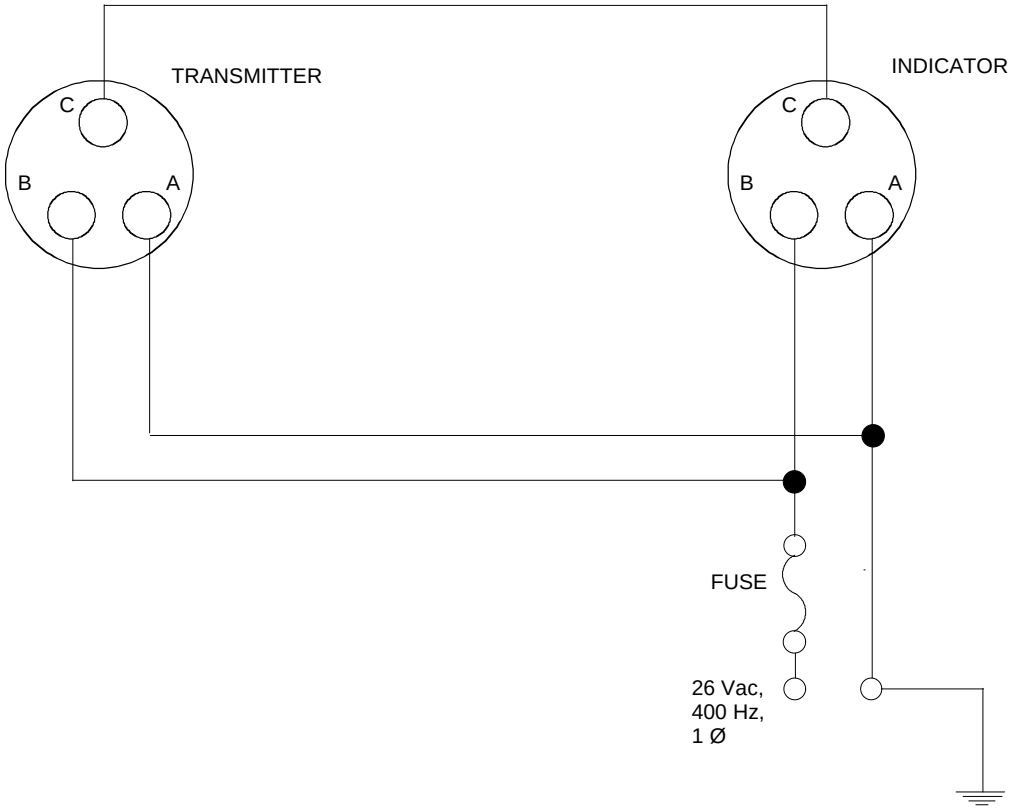
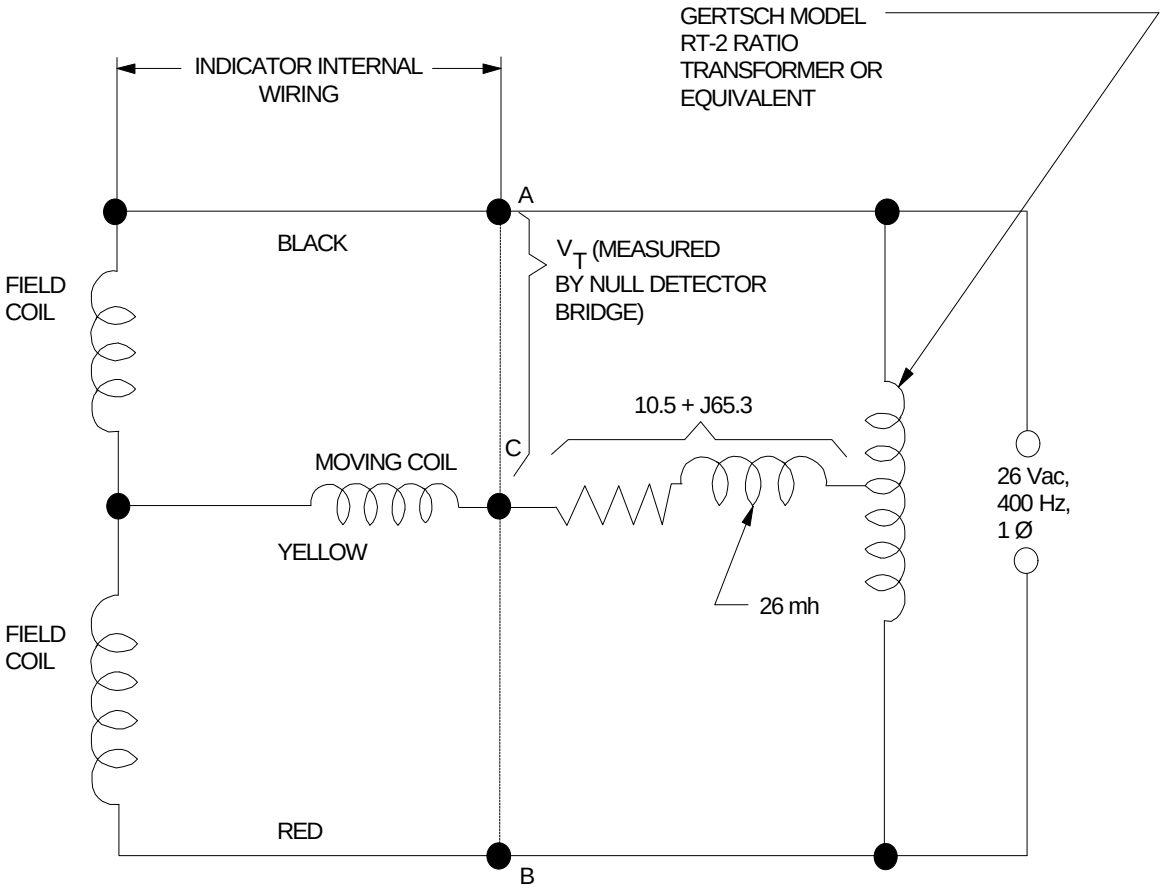


FIGURE 1. External wiring

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INDICATION (% FULL SCALE) = 305 - 19.6 V_T (VOLTS RMS)

FIGURE 2. Test standard circuit

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-38230C	**2. DOCUMENT DATE (YYMMDD)** 970110

3. DOCUMENT TITLE

TRANSMITTER, PRESSURE, OIL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	**d. TELEPHONE** *(Include Area Code)* (1) Commercial (2) Autovon *(if applicable)*	**7. DATE SUBMITTED (YYMMDD)**

8. PREPARING ACTIVITY

a. NAME Harlena Edwards	**b. TELEPHONE** *(Include Area Code)* (1) Commercial (2) Autovon (405) 736-5960 336-5960
c. ADDRESS ***(Include Zip Code)*** OC-ALC/TICLA 3001 Staff Drive, Suite 1AE1-101A Tinker AFB, OK 73145-3036	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3446** **Telephone (703) 756-2340 Autovon 289-2340**

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